FC6 [FC6]

FC6 Properties							
General							
Name	FC6	**Number**	6	**Type**	FC	**Language**	STL
Numbering	Manual						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Offset	Default value	Supervision	Comment
▼ Input					
ACQ	Word				
SS	Real				
SI	Real				
a	Real				
b	Real				
▼ Output					
STAT	Byte				
VAL	Real				
InOut					
Temp					
Constant					
▼ Return					
FC6	Void				

Network 1:

```
0001        L     0                  // Initializations
0002        T     %MB20
0003        T     #STAT
0004        L     #ACQ               // Load acquired value
0005        T     %MW10              // Save in MW10
0006        L     W#16#7FFF          // If acquisition error
0007        ==I                      // Other base
0008        SPB   ESCI               // Return
0009        L     %MW10              // Load acquired value
0010        *ITD                     // Convert to 32-bit
0011        DTR                      // Convert to real
0012        L     #a                 // Load parameter a
0013        *R                       // Multiply
0014        L     #b                 // Load parameter b
0015        +R                       // Sum
0016        T     #VAL               // Copy converted value on output parameter VAL
0017        T     %MD12              // Save converted value in MD12
0018        L     #SS                // Load high threshold
0019        >R                       // Comparison
0020        =     %M20.1             // Copy result of comparison in M20.1
0021        L     #SI                // Load low threashold
0022        L     %MD12              // Load converted value
0023        >R                       // Comparison
0024        =     %M20.2             // Copy result of comparison in M20.2
0025        L     %MB20
0026        L     0
0027        <>I                      // Check if MB20 not 0
0028        =     %M20.7             // Report in cumulative error bit
0029        L     %MB20              // Load status byte
0030        T     #STAT              // Copy on output parameter STAT
0031        BEA                      // Quit or you execute the error branch
0032 ESCI: L     DW#16#FFFFFFFF     // Load error constant
0033        T     #VAL               // Copy on parameter VAL
0034        SET                      // Put RLO =1
0035        =     %M20.0             // Copy to M20.0 and M20.7
0036        =     %M20.7
0037        L     %MB20              // Load status byte
0038        T     #STAT              // Copy on output parameter STAT
0039
0040
0041
0042
0043
0044
0045
0046
0047
0048
0049
0050
0051
0052
0053
0054
0055
```

0056
0057

output parameters ⇒ you need to specify where to write them
⇒ no numeric values as for inputs

e.g. STAT := MB60